STATISTICS WORKSHEET- 6

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1. Which of the following can be considered as random variable?

d) All of the mentioned

2. Which of the following random variable that take on only a countable number of possibilities?

c) Continuous

3. Which of the following function is associated with a continuous random variable?

a) pdf

4. The expected value or _____ of a random variable is the center of its distribution.

c) mean

5. Which of the following of a random variable is not a measure of spread?

c) empirical mean

6. The _____ of the Chi-squared distribution is twice the degrees of freedom.

a) variance

7. The beta distribution is the default prior for parameters between _____

c) 0 and 1

8. Which of the following tool is used for constructing confidence intervals and calculating standard errors for difficult statistics?

b) bootstrap

9. Data that summarize all observations in a category are called _____ data.

a) frequency

10. What is the difference between a boxplot and histogram?

A histogram is a graphical representation of the distribution of a dataset. It shows the frequency or count of observations that fall within a series of intervals, which represents the range of values in the dataset.
The height of each bar or column corresponds to the frequency of observations.
Histograms can provide insights into the shape, central tendency, variability, and skewness of a distribution, as well as any outliers or gaps in the data.

A boxplot is a graphical summary of the distribution of a dataset based on the following values:
the first quartile (25th percentile),
the median (50th percentile),
the third quartile (75th percentile),
and the maximum value (100 percentile).
The box in the middle of the plot represents the interquartile range (IQR), which is the range of values that contains the middle 50% of the data. The whiskers extend from the box to the smallest and largest observations within 1.5 times the IQR from the box. Boxplots can provide insights into the shape, central tendency, variability, and outliers of a distribution, as well as any asymmetry or skewness.

11. How to select metrics?

- Define the problem or question: Clearly define the problem or question.This will help you identify the appropriate metrics to use.

- Identify the type of data: Determine the type of data you are working with, as this will guide the selection of appropriate metrics. Data can be classified as numerical or categorical.

- Choose the appropriate measurement scale: Choose an appropriate measurement scale based on the type of data. For numerical data, there are four measurement scales: nominal, ordinal, interval, and ratio. For categorical data, the measurement scale is usually nominal.

- Identify relevant variables: Identify the variables that are relevant to the problem or question. These may be dependent or independent variables.

- Choose appropriate metrics: Choose appropriate metrics to measure the variables of interest. Some common metrics include measures of central tendency (e.g., mean, median, mode), measures of variability (e.g., standard deviation, variance), correlation coefficients, and regression coefficients.

- Consider the context: Consider the context in which the data was collected and the specific goals of the analysis. This may influence the choice of metrics.

- Evaluate the quality of metrics: Evaluate the quality of metrics to ensure they are valid, reliable, and sensitive to the data.

12. How do you assess the statistical significance of an insight?

- Define the hypothesis: Start by defining a null hypothesis and an alternative hypothesis. The null hypothesis is typically a statement of no effect, while the alternative hypothesis is the opposite of the null hypothesis

- Choose a statistical test: Choose an appropriate statistical test based on the type of data and the hypothesis being tested. For example, if the data is normally distributed, and you want to compare means between two groups, you may choose a t-test. If the data is not normally distributed, you may choose a non-parametric test like the Mann-Whitney U test.

- Set a significance level: Choose a significance level, which is typically denoted by alpha (α). This represents the maximum probability of rejecting the null hypothesis when it is true. The most commonly used significance level is 0.05, which means there is a 5% chance of rejecting the null hypothesis when it is actually true.

- Calculate the test statistic: Calculate the test statistic based on the chosen statistical test and the data.

- Determine the p-value: Determine the p-value, which is the probability of obtaining a test statistic as extreme as, or more extreme than, the observed value, assuming the null hypothesis is true. A p-value less than or equal to the significance level indicates that the observed result is statistically significant, and the null hypothesis can be rejected in favour of the alternative hypothesis.

- Interpret the results: Based on the p-value and the significance level, interpret the results and draw conclusions about the statistical significance of the insight.

13. Give examples of data that does not have a Gaussian distribution, nor log-normal.

Poisson Distribution: This is a discrete probability distribution that is often used to model count data. It describes the probability of a certain number of events occurring in a fixed interval of time or space. Examples of data that follow a Poisson distribution include the number of customers who visit a store in a day or the number of car accidents that occur on a particular road in a month

Gamma Distribution: This is a continuous probability distribution that is often used to model data that are positive and skewed. Examples of data that follow a Gamma distribution include the time it takes for a machine to break down or the amount of rainfall in a particular area over a period of time

Exponential Distribution: This is a continuous probability distribution that is often used to model the time between two successive events in a Poisson process. Examples of data that follow an Exponential distribution include the time between customer arrivals at a store or the time between earthquakes in a particular region

Binomial Distribution: This is a discrete probability distribution that is often used to model data that have two possible outcomes. Examples of data that follow a Binomial distribution include the number of heads obtained in a series of coin flips or the number of defective products produced in a manufacturing process.

14. Give an example where the median is a better measure than the mean.

Suppose in a class, there are 5 students and their marks out of 10 are as follows:
1, 1, 1, 10, 10 respectively.

The mean marks for these students is (1+1+1+10+10)/5 = 4.6
This is a misleading measure of central tendency because the two highest marks are outliers that are significantly larger than the other marks.
The median marks, on the other hand, is the middle value when the marks are arranged in order: 1.
This provides a more accurate representation of the typical marks in this class, as it is not influenced by the extreme outliers.
In this example, the median is a better measure than the mean because the mean is heavily influenced by the outliers, while the median provides a more representative value of the central tendency of the data.

15. What is the Likelihood?

In statistics, likelihood refers to the probability of observing a particular set of data given a specific set of parameter values in a statistical model.

The likelihood function is a function of the parameters of the statistical model and is used to estimate the most likely values for those parameters given the observed data.

For example, let's say we have a simple statistical model that describes the probability of flipping a coin and getting heads. The model assumes that the coin is fair, meaning that the probability of getting heads is 0.5. If we observe a sequence of coin flips and want to estimate the most likely value of the parameter (i.e., the probability of getting heads), we can use the likelihood function.